

Farzaneh Ghanbari

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RESEARCH INTERESTS

Knowledge tracing and student modeling; interpretability and faithfulness of deep sequential models; educational data mining; the use of large language models to produce verifiable, teacher-facing explanations of model predictions.

EDUCATION

Ph.D. in Computer Science 2025 – present

New Mexico State University, Las Cruces, NM, USA

- Advisor: Dr. Christabel Wayllace
- Dissertation area: interpretability and faithfulness of knowledge tracing models
- Cumulative GPA: 4.00

M.Sc. in Computer Science (Scientific Computing) 2013 – 2016

Tarbiat Modares University, Tehran, Iran

- Thesis: a face recognition method based on sparse representation
- Supervisor: Prof. Mansour Rezghi
- Coursework and topics: sparse representation, classification, clustering algorithms, data mining

B.Sc. in Computer Science 2008 – 2012

Shahid Bahonar University of Kerman, Kerman, Iran

- Thesis: analysis, design and implementation of a hotel booking system using a software engineering approach
- Supervisor: Prof. Mohamadreza Omid

RESEARCH EXPERIENCE

Doctoral Research — Knowledge Tracing Interpretability

2025 – present

Department of Computer Science, New Mexico State University

- Evaluating whether knowledge tracing models that claim interpretability — in particular QIKT (AAAI 2023) and IKT (AAAI 2022) — are faithful at the level of individual students, using the pykt-toolkit benchmarking framework on the ASSIST2009 dataset.
- Designed and ran three original experimental studies: cold-start behavior of the model's interpretability scores, the empirical semantics of its three internal prediction streams, and assumption-breaking tests based on controlled edits to student response sequences.
- Showed that the model's named interpretability scores are largely redundant, and that its explanations are unstable during a student's early interactions even though predictive accuracy (AUC) remains stable.
- Built a pipeline that generates teacher-facing explanations from the model's internal scores and validates LLM-generated explanations through hallucination checks, faithfulness grading against measured correlation ceilings, and forbidden-claim detection.
- Reproduced the MSKT metacognitive knowledge tracing model on the EdNet-KT4 dataset, obtaining results close to those reported in the original paper.
- Independently identified an implementation inconsistency in the published QIKT source code and overparameterization in one of its modules.
- Experiments run on a multi-GPU Linux cluster (NVIDIA H100) using PyTorch and conda environments.

M.Sc. Research — Sparse Representation for Face Recognition

2013 – 2016

Tarbiat Modares University, Tehran, Iran

- Developed and evaluated a face recognition method based on sparse representation, combining dictionary-based classification with clustering techniques.

TEACHING EXPERIENCE

Graduate Teaching Assistant

Fall 2025 – present

Department of Computer Science, New Mexico State University

- Discrete Mathematics — Fall 2026
- Assembly Language Programming — Spring 2026
- Assembly Language Programming and Operating Systems I — Fall 2025
- Graded assignments and exams, held regular office hours, and assisted students during lab sessions.

OTHER PROFESSIONAL EXPERIENCE

Data Collection Assistant

2025

Department of Mathematics, New Mexico State University

PUBLICATIONS

No peer-reviewed publications to date.

RESEARCH SOFTWARE AND OTHER INTELLECTUAL PRODUCTS

- Evaluation code built on the pykt-toolkit for analyzing QIKT interpretability: cold-start analysis, stream-semantics analysis, and sequence-edit experiments. *(Not yet publicly released.)*
- A pipeline for generating and validating LLM-based teacher-facing explanations from knowledge tracing model outputs.
- Reproduction of the MSKT model on EdNet-KT4.
- Qualifying exam report and presentation analyzing and comparing the IKT (AAAI 2022) and QIKT (AAAI 2023) papers.

TECHNICAL SKILLS

Primary languages	Python, C, C++, Java, MATLAB
Machine learning	PyTorch, pykt-toolkit, scikit-learn, NumPy, pandas
Model families	CNNs, RNNs and LSTMs, Transformers, GANs, diffusion models, graph neural networks
Tools & environments	Linux, conda, Git and GitHub, Docker, GPU/HPC clusters (NVIDIA H100), LaTeX
Databases	MySQL, SQL Server
Web	HTML5, CSS3, JavaScript
Languages	Persian (native), English

Last updated: September 2026